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# MySQL FOREIGN KEY Constraint

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## MySQL FOREIGN KEY Constraint

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The `FOREIGN KEY` constraint establishes a link between two tables, and prevents action that will destroy the link between them.

A `FOREIGN KEY` is a column in a table that refers to the `PRIMARY KEY` in another table.

The table with the foreign key column is called the child table, and the table with the primary key column is called the referenced or parent table.

The `FOREIGN KEY` constraint prevents invalid data from being inserted into the foreign key column (in the child table), because the value has to exist in the parent table.

The `FOREIGN KEY` constraint also prevents you from deleting a record in the parent table, if related rows still exist in the child table.

Assume we have two tables:

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| PersonID | LastName | FirstName | Age |
|---|---|---|---|
| 1 | Hansen | Ola | 30 |
| 2 | Svendson | Tove | 23 |

## Orders Table

| OrderID | OrderNumber | PersonID |
|---|---|---|
| 3 | 22456 | 2 |
| 4 | 24562 | 1 |

Here we see that the "PersonID" column in the "Orders" table points to the "PersonID" column in the "Persons" table.

The "PersonID" column in the "Persons" table is the `PRIMARY KEY` in the "Persons" table.

The "PersonID" column in the "Orders" table is the `FOREIGN KEY` in the "Orders" table.

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of the "Orders" table:

```
CREATE TABLE Orders (
    OrderID int PRIMARY KEY,
    OrderNumber int NOT NULL,
    PersonID int,
    CONSTRAINT fk_Person
    FOREIGN KEY (PersonID)
    REFERENCES Persons(PersonID)
);
```

# FOREIGN KEY on ALTER TABLE

To create a `FOREIGN KEY` constraint on the "PersonID" column after the "Orders" table is created, use the following SQL:

```
ALTER TABLE Orders
ADD CONSTRAINT fk_Person
FOREIGN KEY (PersonID)
REFERENCES Persons(PersonID);
```

# Drop a FOREIGN KEY Constraint

To drop a `FOREIGN KEY` constraint, use the following SQL:

```
ALTER TABLE Orders
```

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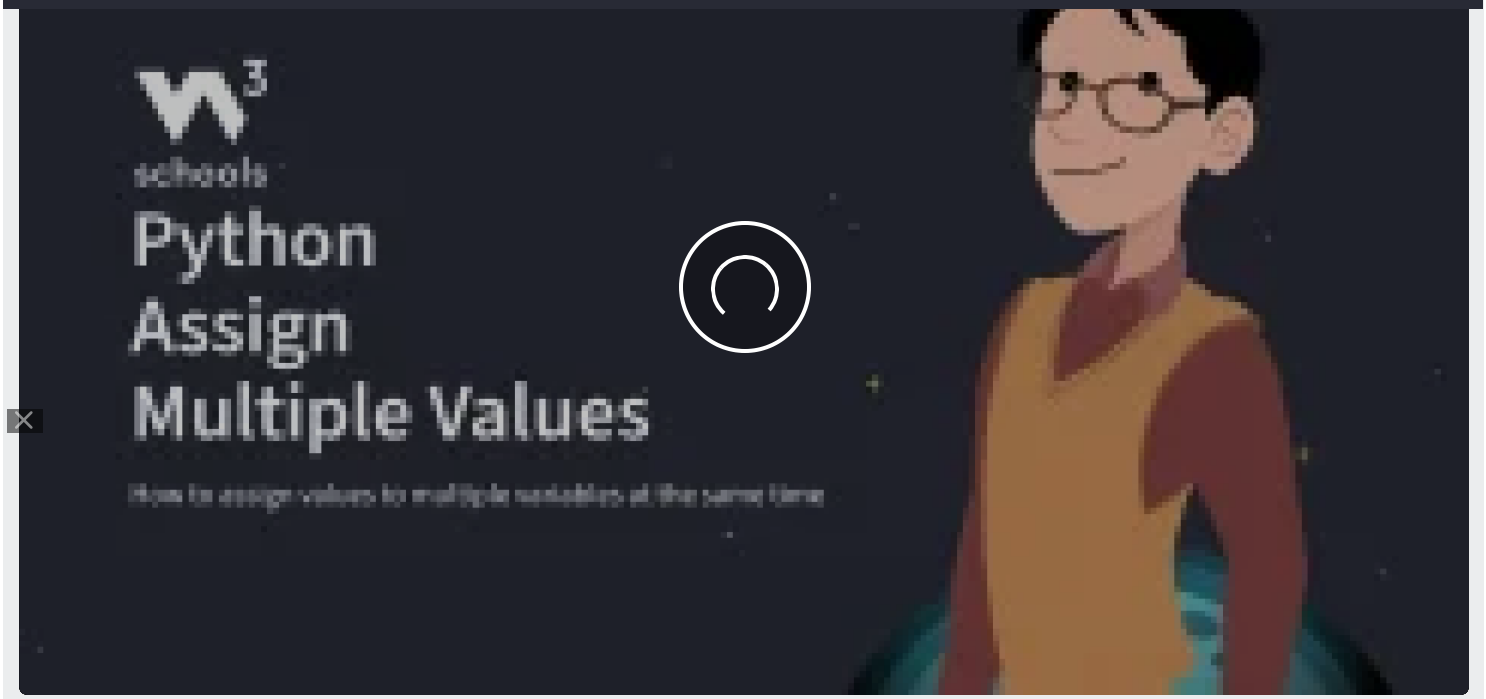
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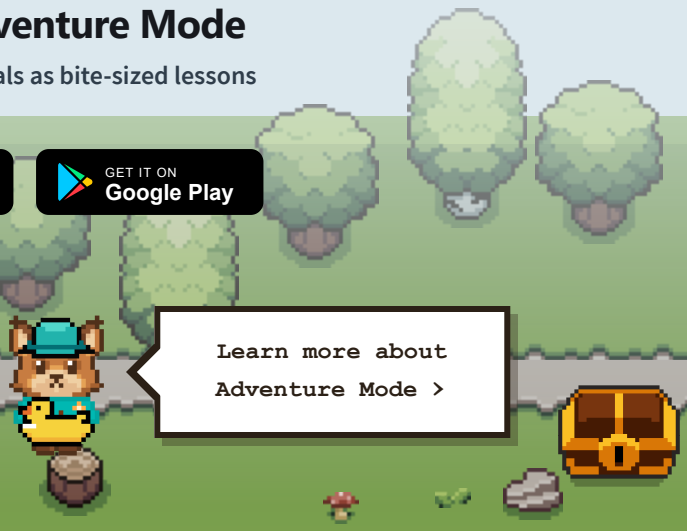
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